

Synoptic CB: Porewater DOC, TN

November 2022 Samples

2026-06-17

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##Setup - Change things here & write any notes

```
#identify section
cat("Setup Information")
```

Setup Information

```
##### Run information - PLEASE CHANGE
Date_Run = "11/19/2022" #Date that instrument was run
Run_by = "Stephanie J. Wilson" #Instrument user
Script_run_by = "Zoe Read" #Code user
run_notes = "Standard Curve assessed manually on the instrument. Some sample IDs are missing from metadata"
samples <- c("GCrew", "GWI", "MSM", "SWH") #whatever identifies your samples within the same names

samples_pattern <- paste(samples, collapse = "|")
#samples_pattern <- "GCW" #use this instead of the line above if you have only one site code
chks_name = "Chk" #what did you name your check standards?

##### File Names - PLEASE CHANGE
#file path and name for raw summary data file
raw_file_name = c("Raw Data/TOCTN_COMPASS_Synoptic_TCTN_202211_1.txt") # , "Raw Data/TOCTN_COMPASS_
```

```

#file path and name for raw all peaks file
# raw_allpeaks_name = "Raw Data/COMPASS_SynopticCB_PW_DOC_202505_allpeaks.txt"

#file path and name of processed data file
processed_file_name = "Processed Data/COMPASS_SynopticCB_PW_Processed_DOC_202211.csv"

##### Log Files - PLEASE CHECK
#downloaded metadata csv - downloaded from Google drive as csv for this year
Raw_Metadata = "Raw Data/COMPASS_SynopticCB_PW_SampleLog_2022.csv"

#qac log file path for this year
#Log_path = "Raw Data/COMPASS_Synoptic_TOCTN_QAClog_2022.csv"

```

```
##Set Up Code
```

```
##Read in metadata and create similar sample IDs for matching to samples
```

0.1 Import Data Functions

0.2 Import Sample Data

```
## Import Sample Data
```

```
## New names:
```

```
## * ' ' -> '...14'
```

```
## # A tibble: 6 x 4
```

```
##   sample_name      npoc_raw tdn_raw run_datetime
##   <chr>          <dbl>    <dbl> <chr>
## 1 1_MSM_202211_UP_LysA_10cm    32.8    1.44 11/19/2022 4:27:48 AM
## 2 2_MSM_202211_UP_LysA_20cm    47.5    2.30 11/19/2022 4:55:54 AM
## 3 3_MSM_202211_UP_LysB_10cm    39.4    0.689 11/19/2022 5:21:49 AM
## 4 4_MSM_202211_UP_LysB_20cm    57.7    1.31 11/19/2022 5:47:57 AM
## 5 5_MSM_202211_UP_LysB_45cm    33.3    1.10 11/19/2022 6:15:47 AM
## 6 6_MSM_202211_UP_LysC_10cm    44.8    1.07 11/19/2022 6:44:17 AM
```

```
##Fix sample naming
```

```

#remove number from sample ID
dat_raw <- dat_raw %>%
  mutate(sample_name = str_remove(sample_name, "^[^_]*_"))

#fix sample missign underscore
dat_raw <- dat_raw %>%
  mutate(sample_name = gsub("LysB20cm", "LysB_20cm", sample_name))

#want date to be at the beginning for now
#MSM_202211_UP_LysA_10cm --> 202209_GWI_UP_LysA_10cm
dat_raw <- dat_raw %>%
  separate(sample_name, into = c("Site", "Date", "Zone", "Replicate", "Depth")) %>%

```

```
mutate(sample_name = paste(Date, Site, Zone, Replicate, Depth, sep = "_")) %>%
mutate(sample_name = gsub("_NA", "", sample_name)) %>%
select(-c("Site", "Date", "Zone", "Replicate", "Depth"))
```

```
## Warning: Expected 5 pieces. Additional pieces discarded in 6 rows [11, 22, 35, 44, 55,
## 66].
```

```
## Warning: Expected 5 pieces. Missing pieces filled with 'NA' in 9 rows [25, 26, 27, 50,
## 51, 52, 77, 78, 79].
```

```
##fix GCW designation
dat_raw <- dat_raw %>%
  mutate(sample_name = gsub("GCrew", "GCW", sample_name)) %>%
  mutate(sample_name = gsub("Gcrew", "GCW", sample_name))
```

0.3 Assessing Standard Curves - assessed manually on the instrument

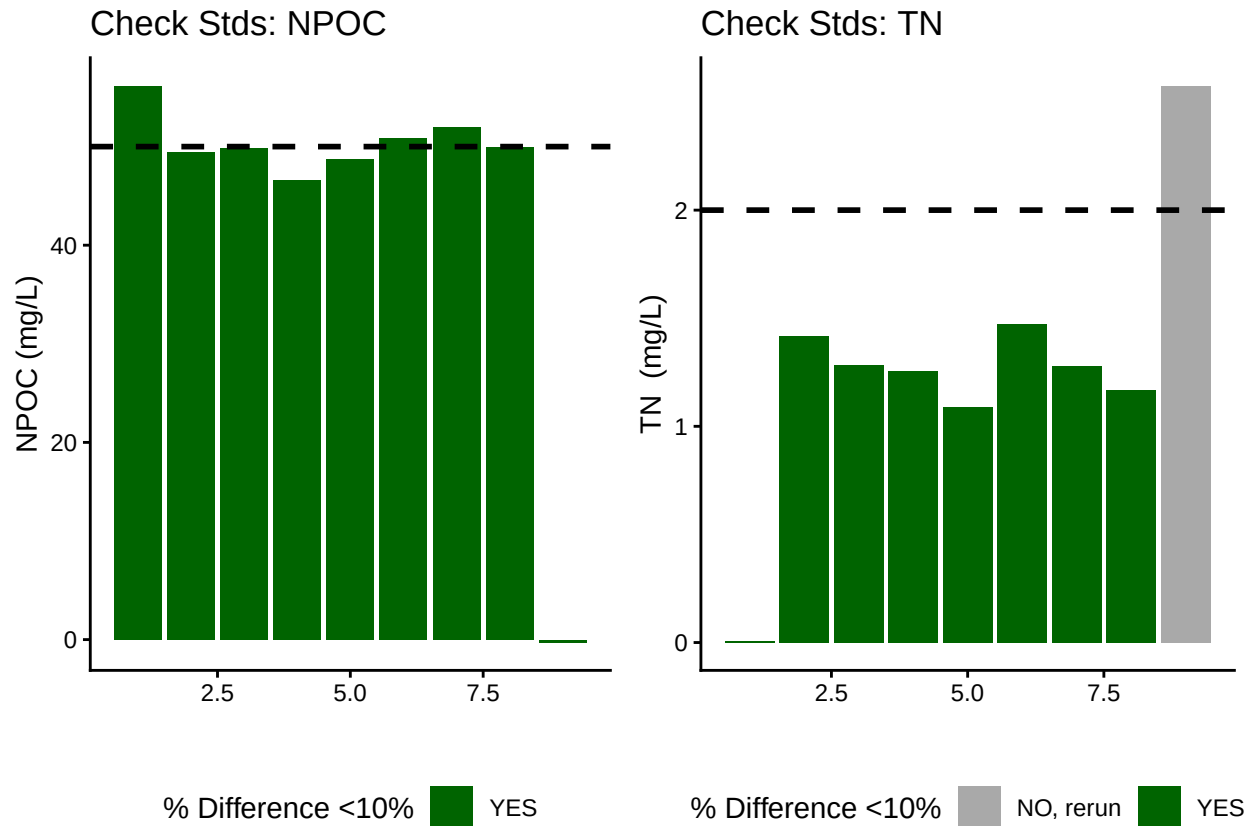
0.4 Assess Check Standards

```
## Assess the Check Standards
```

```
## New names:
## * ' ' -> '...14'
```

```
## [1] "Carbon CHECK STANDARD RSD TOO HIGH - REASSESS"
```

```
## [1] "Nitrogen CHECK STANDARD RSD TOO HIGH - REASSESS"
```



```
## [1] ">50% of Carbon Check Standards are within range of expected concentration"
```

```
## [1] ">50% of Nitrogen Check Standards are within range of expected concentration"
```

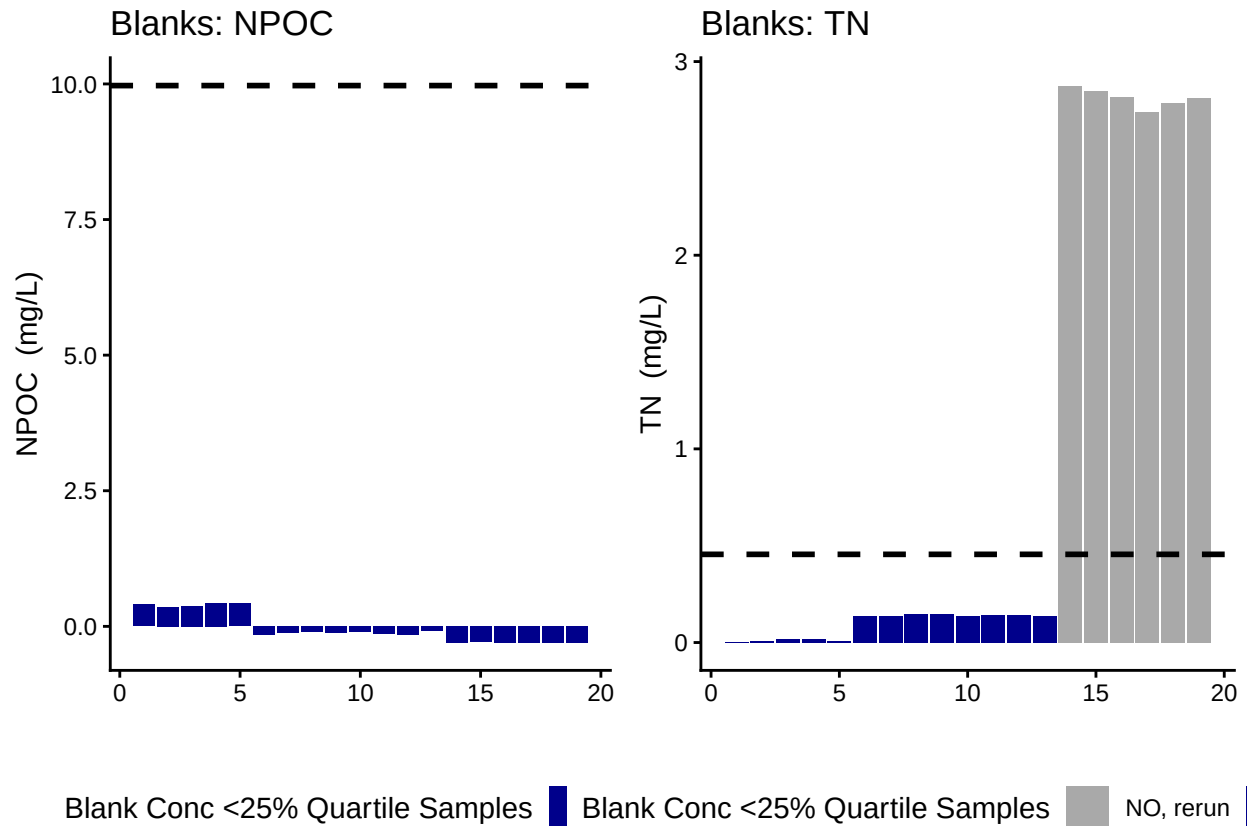
0.5 Assess Blanks

```
## Assess Blanks
```

```
## New names:
## * ' ' -> '...14'
```

```
## [1] ">50% of Carbon Blank concentrations are lower 25% quartile of samples"
```

```
## [1] ">50% of Nitrogen Blank concentrations are lower 25% quartile of samples"
```



```
## carbon blanks:
```

```
## [1] -0.03875105
```

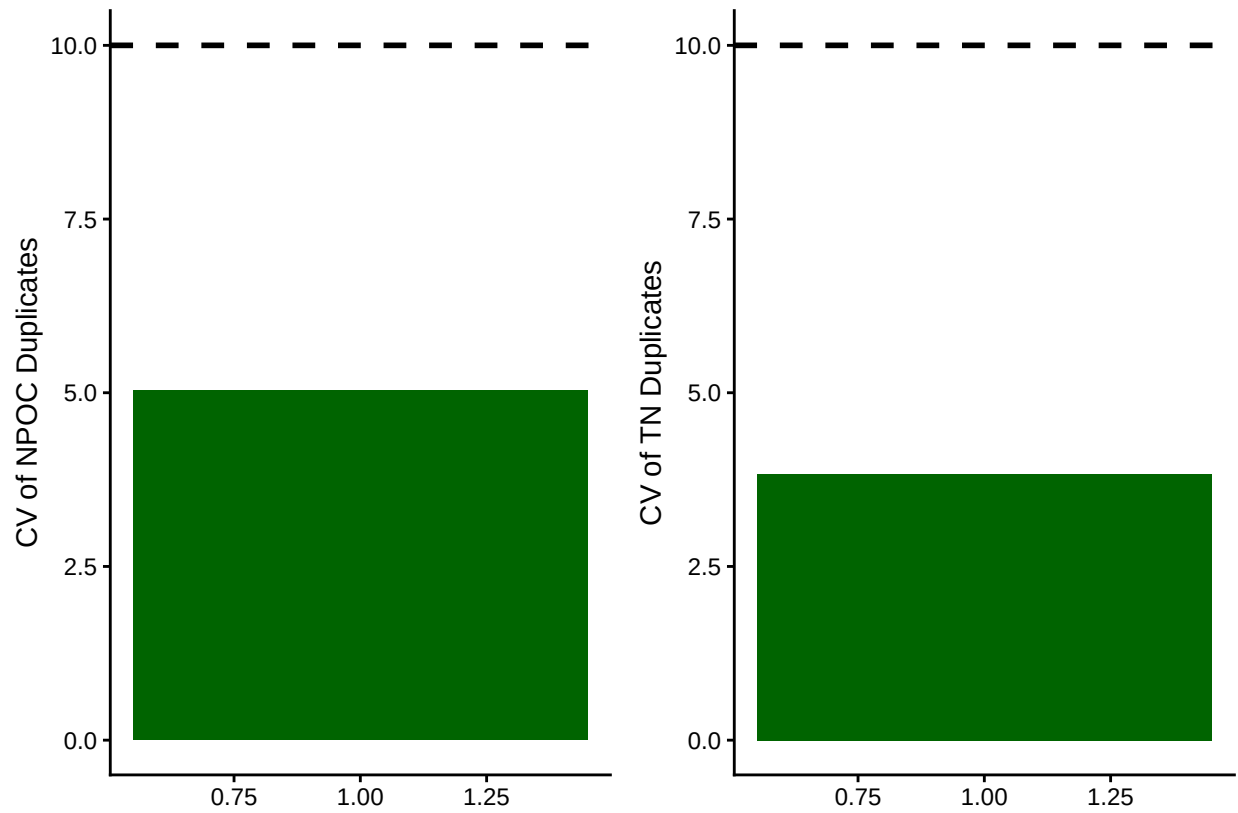
```
## nitrogen blanks:
```

```
## [1] 0.9508305
```

0.6 Assess Duplicates

```
## Assess Duplicates
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

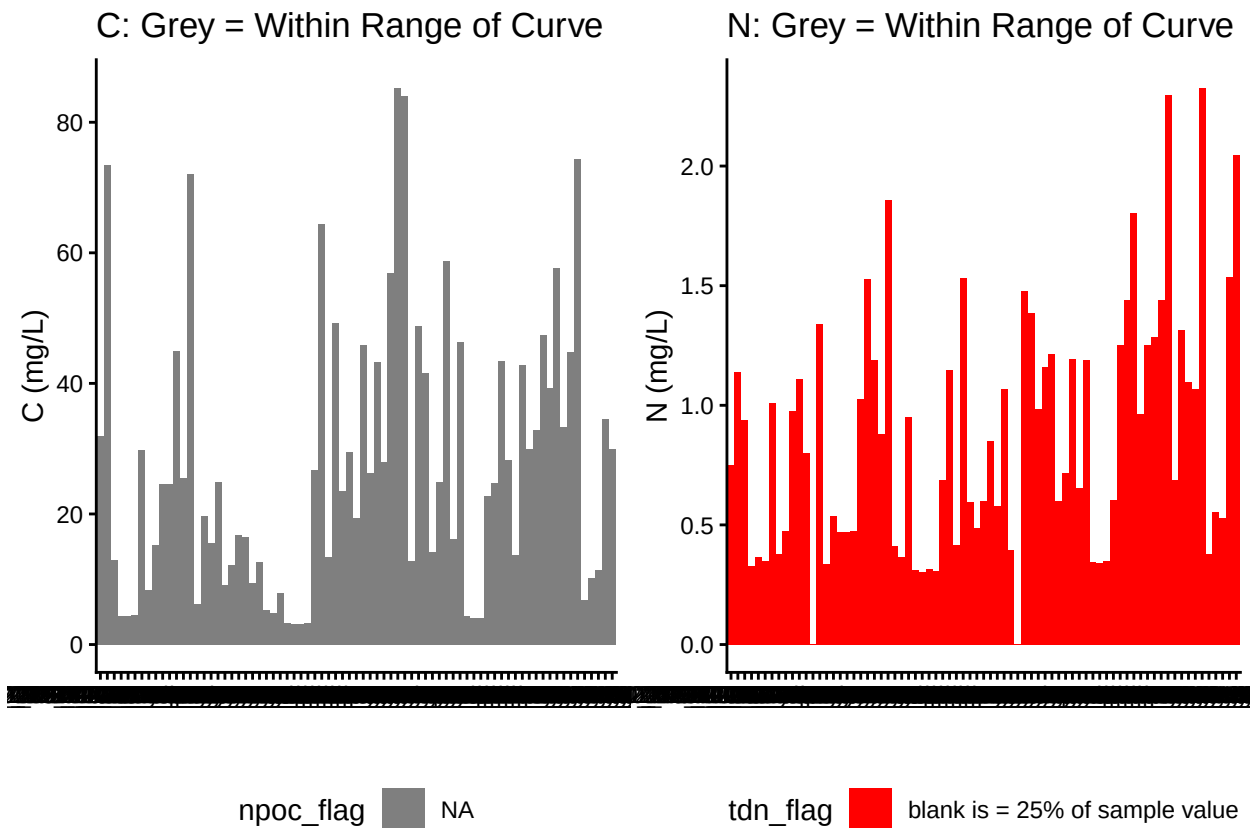


```
## [1] ">50% of Carbon Duplicates have a CV <10%"
```

```
## [1] ">50% of Nitrogen Duplicates have a CV <10%"
```

0.7 Sample Flagging - Are samples Within the range of the curve?

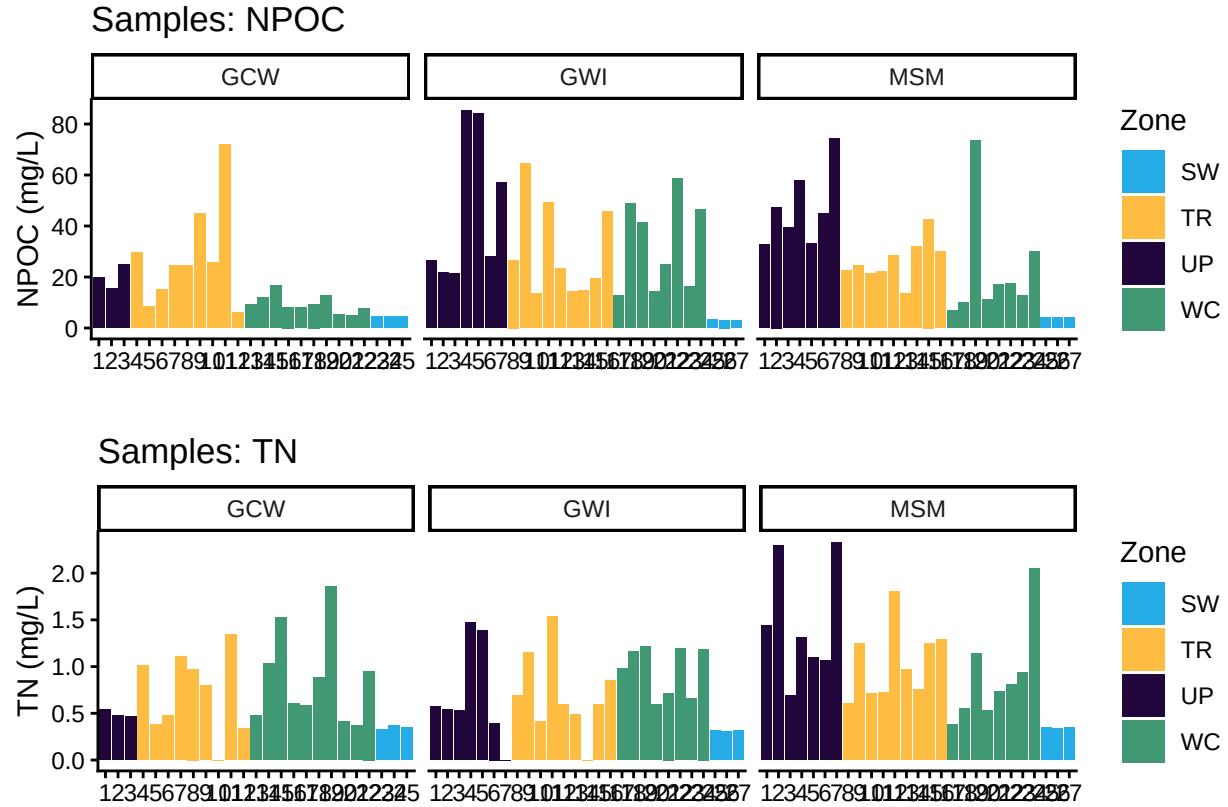
```
## Sample Flagging
```



0.8 Visualize Data by Plot

```
## Visualize Data
```

```
## Warning in rbind(c("202211", "MSM", "UP", "LysA", "10cm"), c("202211", "MSM", :
## number of columns of result is not a multiple of vector length (arg 25)
```



0.9 Convert data from mg/L to uMoles/L

0.10 Check to see if samples run match metadata & merge info

```
## Check Sample IDs with Metadata
```

```
## Some sample IDs are missing from metadata.
```

```
## [1] "202211_MSM_UP_LysA_10cm" "202211_MSM_UP_LysA_20cm"
## [3] "202211_MSM_UP_LysB_10cm" "202211_MSM_UP_LysB_20cm"
## [5] "202211_MSM_UP_LysB_45cm" "202211_MSM_UP_LysC_10cm"
## [7] "202211_MSM_UP_LysC_20cm" "202211_MSM_TR_LysB_10cm"
## [9] "202211_MSM_TR_LysB_20cm" "202211_MSM_TR_LysB_45cm"
## [11] "20211_MSM_TR_LysC_10cm" "202211_MSM_TR_LysC_20cm"
## [13] "202211_MSM_TR_LysC_45cm" "202211_MSM_WC_SipA_10cm"
## [15] "202211_MSM_WC_SipA_20cm" "20211_MSM_WC_SipA_45cm"
## [17] "202211_MSM_WC_SipB_10cm" "202211_MSM_WC_SipB_20cm"
## [19] "20211_MSM_WC_SipC_10cm" "202211_MSM_WC_SipC_20cm"
## [21] "202211_MSM_SW_A" "202211_MSM_SW_B"
## [23] "202211_MSM_SW_C" "202211_GCW_UP_LysB_10cm"
## [25] "202211_GCW_UP_LysB_20cm" "202211_GCW_UP_LysC_10cm"
## [27] "202211_GCW_TR_LysA_20cm" "202211_GCW_TR_LysA_45cm"
## [29] "202211_GCW_TR_LysB_10cm" "202211_GCW_TR_LysB_20cm"
## [31] "202211_GCW_TR_LysB_20cm" "202211_GCW_TR_LysB_45cm"
```



```
## [33] "202211_GCW_TR_LysC_10cm" "202211_GCW_TR_LysC_20cm"
## [35] "202211_GCW_TR_LysC_45cm" "202211_GCW_WC_SipA_10cm"
## [37] "202211_GCW_WC_SipA_20cm" "202211_GCW_WC_SipA_45cm"
## [39] "202211_GCW_WC_SipB_10cm" "202211_GCW_WC_SipB_20cm"
## [41] "202211_GCW_WC_SipB_45cm" "202211_GCW_WC_SipC_10cm"
## [43] "202211_GCW_WC_SipC_20cm" "202211_GCW_WC_SipC_45cm"
## [45] "202211_GCW_SW_A"          "202211_GCW_SW_B"
## [47] "202211_GCW_SW_C"          "202211_GWI_UP_LysA_45cm"
## [49] "202211_GWI_TR_LysB_20cm" "202211_GWI_TR_LysC_45cm"
## [51] "202211_GWI_WC_SipA_10cm" "202211_GWI_WC_SipA_20cm"
## [53] "202211_GWI_WC_SipA_45cm" "202211_GWI_WC_SipB_10cm"
## [55] "202211_GWI_WC_SipB_20cm" "202211_GWI_WC_SipB_45cm"
## [57] "202211_GWI_WC_SipC_10cm" "202211_GWI_WC_SipC_20cm"
```

0.11 Export Processed Data

```
## Export Processed Data
```

```
## # A tibble: 6 x 21
##   Project      Region Site Zone Replicate Depth_cm Sample_ID Year Month Day
##   <chr>        <chr> <chr> <chr> <chr>      <chr>      <chr>    <int> <int> <int>
## 1 COMPASS: Sy~ CB    MSM  UP    LysA      10      202211_M~    NA    NA    NA
## 2 COMPASS: Sy~ CB    MSM  UP    LysA      20      202211_M~    NA    NA    NA
## 3 COMPASS: Sy~ CB    MSM  UP    LysB      10      202211_M~    NA    NA    NA
## 4 COMPASS: Sy~ CB    MSM  UP    LysB      20      202211_M~    NA    NA    NA
## 5 COMPASS: Sy~ CB    MSM  UP    LysB      45      202211_M~    NA    NA    NA
## 6 COMPASS: Sy~ CB    MSM  UP    LysC      10      202211_M~    NA    NA    NA
## # i 11 more variables: Time <lgl>, Time_Zone <lgl>, npoc_mgL <dbl>,
## #   npoc_uM <dbl>, npoc_flag <chr>, tdn_mgL <dbl>, tdn_uM <dbl>,
## #   tdn_flag <chr>, Analysis_runtime <chr>, Run_notes <chr>, Field_notes <chr>
```

```
#end
```